

Exam 1 Review 2 DE

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1 Introductory Remarks

These reviews are designed to reflect both the format and difficulty of Professor Ambrose's exams. I encourage you to use them as a guide for identifying the material you should focus on when preparing for each exam.

Differential Equations is a difficult class and requires cumulative preparation. Please note that these packets are only meant to supplement your studying. They do not replace the valuable learning that comes from attending lecture or working through the homework.

Solution manuals are provided for each review.

2 Problems

2.1 Problem 1

Solve for y

$$\frac{dy}{dx} = \frac{xy^3}{\sqrt{1+x^2}}$$

2.2 Problem 2

Solve the DE

$$y' = \frac{2y^4 + x^4}{xy^3}$$

2.3 Problem 3

EXACT DE

$$y' = \frac{2 + ye^{xy}}{2y - xe^{xy}}$$

2.4 Problem 4

LINEAR DE

$$xy' = 3y + x^2$$

2.5 Problem 5

LINEAR DE

$$\sin(x)y' + \cos(x)y = 2x$$

2.6 Problem 6

A 300-gallon tank initially contains 120 gallons of pure water. Starting at $t = 0$, salt water containing 5.0 pounds of slater per gallon falls into the the tank at 4 gal/min. The thoroughly mixed solution flows out at 2 gal/min. How much salt is in the tank when the tank overflows?

2.7 Problem 7

At $t = 0$, a 17-pound object is pushed beneath the surface of a calm lake and then released from rest. The object has a buoyancy force of 21 pounds, so it begins to rise after release. The resistance of the water (in pounds) is given by $\frac{v^2}{4}$, where v is the upward velocity of the object in feet per second. If the object reaches the surface in 7 seconds, what is the velocity of the object when it reaches the surface of the lake?